

BACHELOR WITH COMPUTER APPLICATIONS AS MAJOR (CT-2)

6th SEMESTER

CAP622J2: COMPUTER APPLICATIONS _ COMPUTER ORGANIZATION & ARCHITECTURE

CREDITS: THEORY (4) PRACTICAL (2)

COURSE LEARNING OUTCOMES:

After the course the students are expected to be able to:

- *Explain design of the various functional units and components of computers.*
- *Explain the basics of organizational and architectural issues of a digital computer and Classify and compute the performance of machines.*
- *Explain principles of computer organization and the basic architectural concepts.*
- *Demonstrate memory management and I/O techniques.*
- *To have better idea on how to write assemble language programs.*
- *Explain key aspects of Computer Organization & Architecture by enabling them to perform the experiments with support of a design and simulation.*

UNIT 1: BASIC COMPUTER ORGANISATION (15 HOURS)

Digital Computers, Von Neumann Computer, Basic organisation of a Computer, Computer Generations, Instruction Codes, Stored Program Organization, Computer registers, Computer instructions, Instruction cycle, Fetch and Decode, Common Bus System, Bus Interconnection Structures

UNIT 2: ARITHMETIC LOGIC UNIT (15 HOURS)

General Register Organisation, Control Word, Stack Organisation, Instruction Formats, Addressing Modes (Direct, Indirect, Register, Indexed), Reduced Instruction Set Computer (RISC), Complex Instruction Set Computer (CISC), Instruction Pipelining, Control Unit Implementation: Hardwired and Microprogrammed, Control Memory.

UNIT 3: MEMORY ORGANIZATION (15 HOURS)

Memory Hierarchy, Memory Technology, Main memory, RAM and ROM Chips, Memory Address Map, Memory Connection to CPU. RAM Technologies, SRAM, DRAM, Read Only Memory(ROM), PROM, EEPROM, Flash Memory, Auxiliary memory, Magnetic Disks, Magnetic Tape, Cache memory, Cache Memory Mappings (Direct, Associative, Set-Associative)

UNIT 4: INPUT-OUTPUT ORGANIZATION (15 HOURS)

Peripheral Devices, Input-Output Interface, I/O Bus and Interface Modules, Isolated and Memory Mapped I/O, Modes of Transfer, Programmed I/O, Interrupt-Initiated I/O, Direct Memory Access (DMA), Bus Arbitration, DMA Controller, DMA Transfer, Input-Output Processor (IOP). Introduction to parallel Processing, Multi-processing systems. Multicore architecture.

TEXT BOOK:

1. M.Morris Mano-Computer System Architecture, Revised Third Edition, Pearson Education

REFERENCE BOOK

1. William Stallings-Computer Organization and Architecture, Tenth Edition, Pearson Education
2. Carl Hamacher-Computer Organization, Fifth Edition, Tata McGraw Hill
3. Ramesh S Gaonkar-Microprocessor Architecture, Programming, and Applications with the 8085, Prentice Hall

LIST OF PRACTICALS:

Computer Organization and architecture lab consist of performing various experiments in GNU Sim (An open source and platform independent simulator for 8085 microprocessor).

1. Write the working of 8085 simulator GNUsim8085.
2. Write the working of 8086 simulator EMU8086.
3. Discuss basic architecture of 8085
4. Discuss basic architecture of 8086
5. Discuss the various registers available in 8085.
6. Study the complete instruction set of 8085 and write the instructions in the instruction set of 8085 along with examples.
7. Write an assembly language code in GNUsim8085 to demonstrate immediate addressing.
8. Write an assembly language code in GNUsim8085 to demonstrate indirect addressing.
9. Write an assembly language code in GNUsim8085 to demonstrate indexed addressing.
10. Write an assembly language code in GNUsim8085 to implement data transfer instruction.

11. Write an assembly language code in GNUsim8085 to store numbers in reverse order in memory location.
12. Write an assembly language code in GNUsim8085 to implement arithmetic instruction.
13. Write an assembly language code in GNUsim8085 to perform addition of two 8 bit numbers.
14. Write an assembly language code in GNUsim8085 to perform addition of two 16 bit numbers.
15. Write an assembly language code in GNUsim8085 to find 1's & 2's complement of a 8 bit number.
16. Write an assembly language code in GNUsim8085 to add two 8 bit numbers stored in memory and also storing the carry.
17. Write an assembly language code in GNUsim8085 to find the factorial of a number.
18. Write an assembly language code in GNUsim8085 to implement logical instructions.
19. Write an assembly language code in GNUsim8085 to implement stack and branch instructions.
20. Write a program using in GNUsim8085 for Decimal addition and subtraction of two Numbers.
21. Write a program using in GNUsim8085 for Hexadecimal addition and subtraction of two Numbers.
22. Write a program using 8085 Microprocessor for addition and subtraction of two BCD numbers.
23. Write a program to perform multiplication of two 8 bit numbers using 8085.
24. Write a program to perform division of two 8 bit numbers using 8085.
25. Write a program to find the largest and smallest number in an array of data using 8085 instruction set.
26. Write a program to arrange an array of data in ascending and descending order.
27. Write an assembly language code in EMU8086 to store numbers in reverse order in memory location.
28. Write an assembly language code in EMU8086 to implement arithmetic instruction.
29. Write an assembly language code in EMU8086 to perform addition of two 8 bit numbers.
30. Write an assembly language code in EMU8086 to perform addition of two 16 bit numbers.